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Seasonal variations in chemical response to conspecific scents in the spotted spiny lobster, *Panulirus guttatus*

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Abstract Chemical response to conspecific scents mediate aggregations in some species of spiny lobsters, but patterns of co-denning of adult lobsters in their natural habitats vary widely. This is so for *Panulirus guttatus*, a small, sedentary, obligate reef-dwelling species with a protracted, almost year-round reproductive season. We hypothesised that changes in patterns of aggregation may vary with the intensity of reproductive activity in the population, which in turn may cause changes in responses to conspecific scents. To test this hypothesis, we conducted three laboratory experiments with Y-mazes to test for seasonal variations in gender-related and size-related response to conspecific scents in mature *P. guttatus*. When reproductive activity (RA) was high (experiment High-RA, 67.4% reproductive females), males responded significantly to scents of conspecifics of both genders, but females only to the scent of females. When reproductive activity had not yet reached its yearly lowest point (experiment Medium-RA, 50.0% reproductive females), only females responded significantly to scents of other females, and when reproductive activity was at its lowest (experiment Low-RA, 4.8% reproductive females) no treatment yielded significant responses. However, results of the size-related treatments showed that only large individuals responded significantly to scents of other large lobsters during the High-RA experiment, whereas during the Medium-RA only small individuals responded to scents of both small and large lobsters. These results are in accordance with previously observed size-dependent reproduction in *P. guttatus* over the protracted reproductive season. Therefore, chemical response to conspecific scents in *P. guttatus* vary with season, gender, and size and may influence co-denning patterns of mature individuals in the reef habitat.

Keywords chemical cues; *Panulirus guttatus*; reproductive activity; spiny lobster

INTRODUCTION

Conspecific chemical cues play an important role in the social behaviour of many decapods, including spiny lobsters (family Palinuridae) (Dunham 1978; Zimmer-Faust & Spanier 1987; Karavanich & Atema 1998). For example, chemical communication appears to mediate aggregation of conspecifics in *Panulirus interruptus* (Zimmer-Faust et al. 1985), *P. argus* (Ratchford & Eggleston 1998, 2000; Nevitt et al. 2000), and *Jasus edwardsii* (Butler et al. 1999). The influence of chemical cues in the aggregation of conspecifics is consequential because variations in gregariousness may cause marked differences in the patterns of distribution and hence in the susceptibility to the fishery of spiny lobsters (MacDiarmid 1994).

Experiments testing gregarious behaviour and the response of spiny lobsters to conspecific chemical scents (e.g., Childress & Herrnkind 1996; Ratchford & Eggleston 1998, 2000; Butler et al. 1999; Nevitt et al. 2000) have generally used immature individuals as experimental subjects because juveniles of spiny lobsters are highly gregarious (Atema & Cobb 1980). In contrast, aggregations of adult spiny lobsters in their natural habitats vary widely. For example, in California, United States, the distribution of individuals of *P. interruptus* in natural dens was random in the winter but aggregated in the spring, with no relationship between cohabitation

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and lobster gender (Zimmer-Faust & Spanier 1987). In *P. argus*, cohabitation varied greatly among coral reef sites in St. John, United States Virgin Islands, with cohabiting individuals often of the same gender (Herrnkind et al. 1975), whereas aggregations of adult *J. edwardsii* showed clear seasonal trends, which differed between mature males and females (MacDiarmid 1994).

Aggregations of adult *Panulirus guttatus* (Latreille, 1804), a small tropical spiny lobster that occurs throughout the Caribbean Sea, appear similarly variable both in the field and in the laboratory (Sharp et al. 1997; Lozano-Álvarez & Briones-Fourzán 2001; Segura-García et al. 2004; Lozano-Álvarez et al. unpubl. data). Individuals of *P. guttatus* are obligate crevice-dwellers and live in the reef habitat throughout their benthic life. Small juveniles are highly cryptic, but mature individuals are sedentary and have a limited home range (Sharp et al. 1997; Lozano-Álvarez et al. 2002; Negrete-Soto et al. 2002). Reproduction occurs almost year-round and large females breed more often through the protracted reproductive period than small mature females. A period of minimal reproduction occurs around late summer–early autumn, but the onset of this period varies from year to year (Chitty 1973; Briones-Fourzán & Contreras-Ortiz 1999; Negrete-Soto et al. 2002).

During field research into the patterns of den use of individuals of *P. guttatus* in fore-reef habitats in the Mexican Caribbean, we recorded more aggregations of individuals during winter–spring than during summer–autumn (Lozano-Álvarez et al. unpubl. data). We hypothesised that this pattern could be related to changes in the response to conspecific chemical cues in adult *P. guttatus* related to changes in the general reproductive activity of the population. We herein present results on laboratory experiments to test this hypothesis.

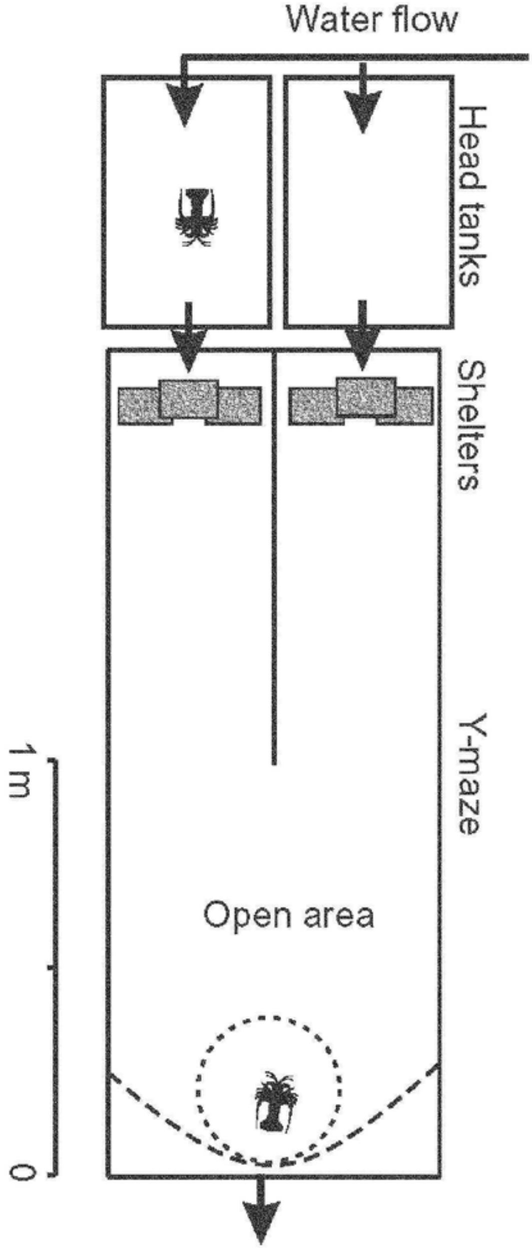

Fig. 1 Schematic representation of an experimental unit to test for response to conspecific scents in *Panulirus guttatus*. Sea water flowed through two head tanks, one of which contained a conspecific (scent producer), then into the arms of the Y-maze, just above two equal shelters made with concrete blocks. Water from both arms mixed in the open area of the Y-maze. One test lobster was placed at the start area of the Y-maze in the early evening and allowed to acclimatise in a wire mesh cylinder, which was removed after 2 h. Test lobster made a shelter choice by the following morning.

METHODS

The experiments were conducted in the Unidad Académica Puerto Morelos, Universidad Nacional Autónoma de México, located at Puerto Morelos on the Caribbean coast of Mexico (20°54′N, 86°54′W). Lobsters were collected by hand from the coral reef habitat, using nocturnal SCUBA diving. A few additional individuals were caught in traps (Segura-García et al. 2004). Lobsters were transported to the laboratory within 1 h of their collection and kept in acclimatisation tanks, 2 m in diameter and 1 m in height, for 2 weeks. The acclimatisation tanks and the experimental units (see below) were kept under shade. A continuous seawater flow maintained ambient temperature in the tanks. Shelter for lobsters in the acclimatisation tanks was provided in the form of multiple segments of dark PVC pipes, 7.3 cm in diameter, sealed on one end and scattered over the floor of the tanks. During the acclimatisation period, lobsters were fed every other day with clams and mussels, but they were not fed during the experiments. Lobsters that moulted during the acclimatisation period were not used in the experiments. At the end of each experiment, lobsters were tagged and released in their natural habitat. Tags were applied to avoid collecting these lobsters again and using them in subsequent experiments.

Experimental set-up

We used four experimental units, each consisting of a Y-maze and two head tanks, to test for chemical response of individuals of *P. guttatus* to conspecific scents (Bushmann & Atema 1997; Ratchford & Eggleston 1998, 2000). The head tanks measured 0.6 × 0.5 × 0.5 m and each held c. 90 litres of water when filled to a standpipe height of 0.3 m. The Y-maze measured 2.0 × 0.8 × 0.6 m, and contained c. 500 litres when filled to a standpipe height of 0.3 m. A panel measuring 1.0 × 0.6 m divided half the length of the Y-maze into two equal arms (Fig. 1). Sea water from a continuous, open system flowed into the head tanks and from each head tank to an arm of the Y-maze. The water then mixed in an open area at the end of the two arms and later flowed out a standpipe at the opposite extreme of the tank, called the "start area" of the maze. The fibreglass head tanks and Y-mazes were completely opaque to preclude visual contact between lobsters and were mounted on different surfaces to eliminate the transference of vibrations. Also, to prevent the transference of acoustic cues through water, the water from the head tanks fell from a height of 5 cm above the water surface of the Y-maze.

Two identical shelters were placed, one in each arm. One of the shelters received water that had flowed through a head tank containing a conspecific whereas the other received water that had flowed through a head tank that held no lobster. The two shelters in each Y-maze were built with three concrete blocks forming a pyramid with a central den and were located 1.8 m from the start area, near the extremes of the Y arms (Fig. 1). Water flow into the head tanks was 1.5–2.0 litre min^{-1}. To prevent the possibility of lobsters using the corners in the start area of the Y-maze as refuges (Ratchford & Eggleston 1998), a semicircular wire-mesh screen was positioned at the start area (Fig. 1).

Trials were conducted at night. Lobsters were randomly assigned to specific treatments and to the left or right head tanks. The lobsters that were held in the head tanks were designated "scent producers" and those tested in the Y-maze "test lobsters". The scent producer was placed in the head tank 30 min before dark, and the test lobster was then placed in the start position of the Y-maze, where it was allowed to acclimatise within a mesh cylinder (0.45 m diam., 0.4 m height). After 2 h, the cylinder was removed and the test lobster was free to roam the Y-maze. Between 0900 and 1000 h the following morning, we recorded the position of the test lobster in the Y-maze as well as the flow rate into the head tanks. All lobsters were then removed and measured (carapace length (CL) from between the rostral horns to the posterior margin of the carapace, ± 0.1 mm). Lobsters were measured at the end of each trial to minimise stress from handling before the trials. To ensure that no scents remained after each trial, the head tanks and Y-mazes were drained and thoroughly brushed. Water was then allowed to flow at a high rate until the beginning of the next trial. No lobster was used more than once as scent producer and no lobster was used more than once as test lobster.

Experimental trials

Our main objective was to explore whether: (1) mature lobsters responded to conspecific scents; (2) this response was gender-related; and (3) the response was related to reproductive activities. Therefore, we used only sexually mature individuals (i.e., females 35 mm CL; males 45 mm CL, Sharp et al. 1997; Briones-Fourzán & Contreras-Ortiz 1999; Robertson & Butler 2003) and conducted three experiments at different seasons reflecting different intensities in reproductive activity in the general population, as indicated by the percentage of all females collected that were carrying egg-masses, new spermatophores, or remnants of egg capsules in their pleopods (reproductive females). We designated these experiments as High-RA (high reproductive activity, 67.4% reproductive females, February–April 2002), Medium-RA (medium reproductive activity, 50.0% reproductive females, July–September 2003), and Low-RA (low reproductive activity, 4.8% reproductive females, July–September 2002). The proportion of reproductive females was significantly different between experiments High-RA and Medium-RA ($\chi^2 = 4.97$, d.f. = 1, $P = 0.025$). Although experiments Medium-RA and Low-RA were conducted over the same period in consecutive years, the different percentages of reproductive females were a result of the annual variation in the onset of the period of minimal reproductive activity.

Each experiment consisted of four treatments in which the paired combinations of scent producers:test lobsters were, respectively, males:males (M:M), females:females (F:F), males:females (M:F), and females:males (F:M). At least 18 replicates were run for each treatment. The result of each treatment was analysed as a one-tailed binomial test (Zar 1999). The null hypothesis was that choosing the shelter having the conspecific scent would have a probability of 0.5.

However, the possibility existed that the release of scent was size-related rather than gender-related (Ratchford & Eggleston 1998); therefore, we reanalysed the results by categorising lobsters into small and large individuals, based on the median size of all our experimental females and males. Thus, small lobsters comprised females 35.0–54.0 mm CL and males 45.0–62.0 mm CL, whereas large lobsters included females 54.1–66.8 mm CL and males 62.1–79.0 mm CL. This procedure yielded four paired combinations of scent producers:test lobsters—small:small (S:S), small:large (S:L), large:small (L:S), and large:large (L:L). The result of each size-related treatment was also analysed as a one-tailed binomial test.

In 55 trials during experiment High-RA, we observed the movement of test lobsters every 30 min during the first 1.5 h after the removal of the mesh cylinders, and recorded their position at the end of this period to assess whether lobsters explored both arms of the Y-maze or whether they chose one arm from the beginning of the experiment.

RESULTS

The movements of the 55 test lobsters observed during experiment High-RA for 1.5 h following the removal of the mesh cylinders were very variable. During this period, most lobsters explored one or both arms of the Y-maze but some remained in the start position or in the open area of the Y-maze. The arm of the Y-maze that the test lobsters first explored did not correspond to the arm containing the conspecific scent (26 of 55 trials) or to the arm containing the shelters chosen by the following morning (26 of 55 trials). Therefore, response of a test lobster to the scent of conspecifics was not immediate.

Significant differences in the mean size of lobsters occurred among experiments (one-way ANOVA on log-transformed data: $F = 18.078$, d.f. = 2, 533; $P < 0.0001$). Lobsters used in experiment High-RA were overall larger (mean ± SD: 59.4 ± 8.3 mm CL) than those used in experiments Medium-RA (54.3 ± 8.4 mm CL) and Low-RA (56.8 ± 7.4 mm CL).

Results of the gender- and size-related treatments are shown in Table 1. Over experiment High-RA, males responded significantly to male and female scents, whereas females only responded significantly to female scents. However, the only size-related treatment with a significant result was that of large individuals responding to scents of other large individuals (Table 1). In experiment Medium-RA, only females responded significantly to female scents, and only small individuals responded significantly to scents of both small and large conspecifics (Table 1). None of the trials conducted during experiment Low-RA yielded significant results (Table 1).

These results suggested that variations in the response to conspecific scents were related to the intensity of reproductive activity, but also that the response was highly influenced by the interaction of gender and size of individuals at different seasons. To fully test this interaction, it would have been necessary to break down the results of each experiment into four combinations of scent producers (SF, LF, SM, LM), each tested against the same four combinations of test lobsters. However, this procedure would have yielded too few replicates in most trial combinations to be of statistical consequence. As an alternative approach, we rearranged the data to test separately the joint effect of gender and size of scent producers on the response of test lobsters of either gender regardless of their size, and the joint effect of gender and size of test lobsters on their response to scent producers of either gender, also irrespective of size. This procedure yielded few replicates in about one third of the "new" treatments, but provided some insight into partial interaction effects (Table 2).

During experiment High-RA, the only new treatments with significant results were those of females responding to scents of large females, and of males responding to scents of large females and large males (Table 2, upper part). However, only the large males and females displayed significant responses (Table 2, lower part). In contrast, during experiment Medium-RA, the only new treatments with significant results were those of females responding to scents of large females, small females, and small males (Table 2, upper part), and of small females responding to female and male scents (Table 2, lower part). No new treatments yielded significant results during experiment Low-RA (Table 2).

Table 1 Results of experiments exploring the response to conspecific scents in *Panulirus guttatus*. Experiment High-RA was conducted during a period of high reproductive activity in the population (February–April 2002); experiment Medium-RA during a period of medium reproductive activity (July–September 2003); experiment Low-RA during a period of minimal reproductive activity (July–September 2002). Four treatments were conducted in each experiment to test for different combinations of gender of scent producers and test lobsters (upper part of table; F, female; M, male). Data were then reanalysed to test for different combinations of size (lower part of table; S, small individuals, females 35.0–54.0 mm carapace length (CL) and males 45.0–62.0 mm CL; L, large individuals, females 54.1–66.8 mm CL and males 62.1–79.0 mm CL). (*N*, number of trials; *N* Yes, trials where test lobsters chose the shelter with the conspecific scent. *P* values were based on one-tailed binomial tests (Zar 1999), where the probability of choosing a shelter having the conspecific scent was 0.5.)

		Experiment conducted during:								
Scent producers	Test lobsters	High-RA *N*	High-RA *N* Yes	High-RA *P*	Medium-RA *N*	Medium-RA *N* Yes	Medium-RA *P*	Low-RA *N*	Low-RA *N* Yes	Low-RA *P*
F	F	23	16	0.046	18	15	0.004	23	13	0.339
F	M	23	18	0.005	17	8	0.500	20	13	0.132
M	F	23	15	0.105	20	14	0.058	20	13	0.132
M	M	38	26	0.017	20	13	0.132	27	16	0.221
S	S	19	10	0.500	29	20	0.031	23	14	0.202
S	L	25	16	0.115	21	12	0.332	26	16	0.163
L	S	24	16	0.076	15	12	0.018	23	15	0.105
L	L	39	33	0.000	10	6	0.377	18	10	0.407

Table 2 Breakdown of results to test the combined effect of gender and size of scent producers on the response of test lobsters (*Panulirus guttatus*) of either gender regardless of their size (upper part of table), and the joint effect of gender and size of test lobsters on their response to scent producers of either gender, also irrespective of size (lower part of table). Statistical analyses and labels as in Table 1.

		Experiment conducted during:								
Scent producers	Test lobsters	High-RA *N*	High-RA *N* Yes	High-RA *P*	Medium-RA *N*	Medium-RA *N* Yes	Medium-RA *P*	Low-RA *N*	Low-RA *N* Yes	Low-RA *P*
SF	F	9	4	0.500	13	10	0.046	12	8	0.194
SF	M	8	6	0.144	12	4	0.194	12	7	0.387
LF	F	14	12	0.006	5	5	0.031	11	5	0.500
LF	M	15	12	0.018	5	3	0.500	8	6	0.144
SM	F	10	6	0.377	14	11	0.029	12	7	0.387
SM	M	15	9	0.304	10	5	0.623	13	8	0.291
LM	F	13	9	0.133	6	4	0.344	8	6	0.145
LM	M	23	17	0.017	10	8	0.055	14	8	0.395
F	SF	9	5	0.500	12	10	0.019	11	8	0.113
F	LF	14	11	0.029	6	5	0.109	12	5	0.387
F	SM	10	7	0.172	7	3	0.500	11	7	0.274
F	LM	13	11	0.011	10	5	0.623	9	6	0.254
M	SF	10	7	0.172	13	10	0.046	14	9	0.212
M	LF	13	8	0.291	7	4	0.500	6	4	0.344
M	SM	15	8	0.500	11	8	0.113	10	5	0.623
M	LM	23	18	0.005	9	5	0.500	17	11	0.166

DISCUSSION

Identifying potential mating partners and competitors is an important aspect of social behaviour in resident lobster populations and conspecific, waterborne odours are likely to convey this information (Atema & Cowan 1986; Karavanich & Atema 1998; MacDiarmid & Butler 1999). Locating appropriate shelters is important to spiny lobsters, and conspecific odours may provide individuals with a means of assessing the quality of a potential shelter (Nevitt et al. 2000).

Therefore, lobsters may respond to scents of conspecifics of both genders and tend, in general, to do so. Without any considerations to gender or size, response to conspecific scents would have been significant in our three experiments (experiment High-RA: N trials = 107, 75 responded, $P < 0.0001$; Medium-RA: $N = 75$, 50 responded, $P = 0.003$; Low-RA: $N = 90$, 55 responded, $P = 0.022$). However, specific treatments in which lobsters significantly chose shelters with conspecific scents occurred only during experiments High-RA and Medium-RA, and significant responses varied also with gender and size.

We do not know if what changed in time was the scent production, the receptivity to the scent, or both. Ratchford & Eggleston (2000) found that in *P. argus* it was the scent production, not the receptivity to the scent that varied in time, but their experimental individuals were sexually immature. Because all our experimental individuals were adults, our results suggest that changes in chemical responses were related to the time within the protracted reproductive period, the size of individuals involved in reproductive activities, and probably the physiological state of individuals. For example, physiological changes before and during the moult may change the general body odour of individuals (Atema & Cowan 1986). Thus, interpretation of these results should be set in the context of the reproductive and moult cycles of *P. guttatus*.

In the Mexican Caribbean, large females of *P. guttatus* start breeding early and produce several broods during the prolonged reproductive period. In contrast, the onset of both the beginning and the end of reproductive activity in small females varies more widely from year to year, but spring is an important reproductive season for both small and large females (Briones-Fourzán & Contreras-Ortiz 1999). Thus, as expected, significantly more females were reproductive during experiment High-RA than during experiment Medium-RA. However, the percentage of small reproductive females was similar in both experiments (68.6 and 55.3% respectively, $\chi^2 = 1.48$, d.f. = 1, $P = 0.224$), whereas the percentage of large reproductive females was greater in experiment High-RA (66.7%) than in experiment Medium-RA (39.1%) ($\chi^2 = 5.13$, d.f. = 1, $P = 0.024$). This suggests that at the time of experiment Medium-RA, large lobsters were nearer their yearly period of minimal reproductive activity than small lobsters.

We did not moult-stage our experimental lobsters, but most adult *P. guttatus* moult at the end of the reproductive period (Chitty 1973; Robertson & Butler 2003). Moreover, spiny lobsters usually remain secluded during the vulnerable pre- to post-moult stages. If large *P. guttatus* in the population end their reproductive activities and start moulting before small individuals, this could explain the smaller mean size of our experimental lobsters during experiments Medium-RA and Low-RA. This is in accordance with Negrete-Soto et al. (2002), who found that the mean size of trap-caught *P. guttatus* in the reef habitat at Puerto Morelos decreased during June–October.

Inter-gender responses differed between experiments High-RA and Medium-RA. In experiment High-RA, males responded significantly to the scents of large females, but the response of females to male scents was not significant. However, only large males displayed significant responses. Experiment High-RA was conducted at the peak of the reproductive period during which, as noted before, large females of *P. guttatus* may breed several times. Small male spiny lobsters are sperm-limited but large males, capable of repeated copulations (MacDiarmid & Butler 1999), may respond more strongly to the scent of large than small females regardless of ovigerous state because large females may mate again. In contrast, ovigerous females may become less responsive to male scents until after their current eggs hatch. For example, ovigerous females of *J. edwardsii* display no preference for empty or occupied shelters (MacDiarmid & Kittaka 2000).

In experiment Medium-RA, by contrast, inter-gender responses were not significant overall, but small individuals were more responsive to conspecific odours than large individuals. Moreover, it was mostly the small females that responded to male scents, and females in general that responded mostly to scents of small males (Table 2), further supporting the notion that large individuals had already decreased their reproductive activity and that the population in general was approaching the period of minimal reproductive activity. Although female spiny lobsters prefer to mate with large males, they may become less selective when the time of egg extrusion is near (MacDiarmid & Kittaka 2000). We can only speculate that if small females were at this point they would be sensitive to male scents, and more so to scents of small males if scents of large males change when they have exhausted their sperm supply.

In intra-gender treatments, females responded significantly to female scents in experiments High-RA and Medium-RA, but the response of males to male scents differed in both experiments. In *P. guttatus*, cohabitation of multiple females in the same den with one mature male is common (Sharp et al. 1997); thus, the presence of female scent may increase the quality of that shelter to other reproductive females (Nevitt et al. 2000). More large females were reproductive during experiment High-RA than during experiment Medium-RA, hence the differential effect of size on responses shown by females to female scents in both experiments. In the latter experiment, however, the low number of replicates in treatments involving large females may have masked their response to female scents in general.

In contrast to females, males responded significantly to male scents in experiment High-RA but not in experiment Medium-RA, but it was mostly the large males that exhibited a significant response, and trials involving large males were less numerous in experiment Medium-RA. Male spiny lobsters are aggressive towards each other, particularly during the reproductive season (MacDiarmid & Kittaka 2000), and in *P. guttatus* large males confront and are capable of evicting smaller males that approach and enter their den (Lozano-Álvarez & Briones-Fourzán 2001; Segura-García et al. 2004). Therefore, when reproductive activity is high, the presence of male scent may signal to other males the proximity of a suitable den occupied by a potential competitor. This does not mean that males would congregate in a den, only that males respond to scents of other males, because the outcome of their proximity would probably depend on further agonistic interactions (e.g., Atema & Cowan 1986; Karavanich & Atema 1998; Segura-García et al. 2004).

In summary, olfaction appears to play a key role in initiating aggregations of adult *P. guttatus*, particularly during the reproductive period. However, the ability to respond to conspecific scents would not necessarily translate into an indiscriminate aggregation of lobsters, since the outcome of the initial chemical response would depend on the physiological state of individuals and on further behavioural interactions. These results may help to explain seasonal variations in cohabitation patterns of *P. guttatus* (Lozano-Álvarez et al. unpubl. data) and provide an arena for future experiments to test more specific hypotheses.

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REFERENCES

Atema, J.; Cobb, J. S. 1980: Social behavior. *In*: Cobb, J. S.; Phillips, B. F. *ed.* The biology and management of lobsters, Vol 1. New York, Academic Press. Pp. 409–450.

Atema, J.; Cowan, D. F. 1986: Sex-identifying urine and molt signals in lobster (*Homarus americanus*). *Journal of Chemical Ecology 12*: 2065–2080.

Briones-Fourzán, P.; Contreras-Ortiz, G. 1999: Reproduction of the spiny lobster *Panulirus guttatus* (Decapoda: Palinuridae) on the Caribbean coast of Mexico. *Journal of Crustacean Biology 19*: 171–179.

Bushmann, P. J.; Atema, J. 1997: Shelter sharing and chemical courtship signals in the lobster *Homarus americanus*. *Canadian Journal of Fisheries and Aquatic Science 54*: 647–654.

Butler, M. J. IV; MacDiarmid, A. B.; Booth, J. D. 1999: The cause and consequence of ontogenetic changes in social aggregations in New Zealand spiny lobster. *Marine Ecology Progress Series 188*: 179–191.

Childress, M. J.; Herrnkind, W. F. 1996. The ontogeny of social behaviour among juvenile Caribbean spiny lobsters. *Animal Behaviour 51*: 675–687.

Chitty, N. 1973: Aspects of the reproductive biology of the spiny lobster, *Panulirus guttatus* Latreille. Unpublished MSc thesis, University of Miami, Florida, United States.

Dunham, P. J. 1978: Sex pheromones in Crustacea. *Biological Review 53*: 555–583.

Herrnkind, W. F.; VanDerwalker, J. A.; Barr, L. 1975: Population dynamics, ecology and behavior of spiny lobsters, *Panulirus argus*, of St. John, U.S.V.I.: (IV) Habitation, patterns of movement and general behavior. *Natural History Museum of Los Angeles County Science Bulletin 20*: 31–45.

Karavanich, C.; Atema, J. 1998: Olfactory recognition of urine signals in dominance fights between male lobsters, *Homarus americanus*. *Behaviour 135*: 719–730.

Lozano-Álvarez, E.; Briones-Fourzán, P. 2001: Den choice and shelter occupation patterns in two sympatric lobster species, *Panulirus argus* and *Panulirus guttatus*, under experimental conditions. *Marine and Freshwater Research 52*: 1145–1155.

Lozano-Álvarez, E.; Carrasco-Zanini, G.; Briones-Fourzán, P. 2002: Homing and orientation in the spotted spiny lobster, *Panulirus guttatus* (Latreille, 1804), towards a subtidal coral reef habitat. *Crustaceana 75*: 859–873.

MacDiarmid, A. B. 1994: Cohabitation in the spiny lobster *Jasus edwardsii* (Hutton, 1875). *Crustaceana 66*: 341–355.

MacDiarmid, A. B.; Butler, M. J. IV. 1999: Sperm economy and limitation in spiny lobsters. *Behavioral Ecology and Sociobiology 46*: 14–24.

MacDiarmid, A. B.; Kittaka, J. 2000: Breeding. *In*: Phillips, B. F.; Kittaka, J. *ed*. Spiny lobsters: fisheries and culture. 2nd ed. New York, Academic Press. Pp. 485–507.

Negrete-Soto, F.; Lozano-Álvarez, E.; Briones-Fourzán, P. 2002: Population dynamics of the spotted spiny lobster, *Panulirus guttatus*, in a coral reef on the Mexican Caribbean. *Journal of Shellfish Research 21*: 279–288.

Nevitt, G.; Pentcheff, N. D.; Lohmann, K. J.; Zimmer, R. K. 2000: Den selection by the spiny lobster *Panulirus argus*: testing attraction to conspecific odours in the field. *Marine Ecology Progress Series 203*: 225–231.

Ratchford, S. G.; Eggleston, D. B. 1998: Size- and scale-dependent chemical attraction contribute to an ontogenetic shift in sociality. *Animal Behaviour 56*: 1027–1034.

Ratchford, S. G.; Eggleston, D. B. 2000: Temporal shift in the presence of a chemical cue contributes to a diel shift in sociality. *Animal Behaviour 59*: 793–799.

Robertson, D. N.; Butler, M. J. IV. 2003: Growth and size at maturity in the spotted spiny lobster, *Panulirus guttatus*. *Journal of Crustacean Biology 23*: 265–272.

Segura-García, I.; Lozano-Álvarez, E.; Briones-Fourzán, P. 2004: Within-shelter behaviour of the spotted spiny lobster, *Panulirus guttatus* (Latreille), in simulated communal dens: an exploratory study. *Marine and Freshwater Behaviour and Physiology 37*: 17–30.

Sharp, W. C.; Hunt, J. H.; Lyons, W. C. 1997: Life history of the spotted spiny lobster, *Panulirus guttatus*, an obligate reef-dweller. *Marine and Freshwater Research 48*: 687–698.

Zar, J. H. 1999: Biostatistical analysis. 4th ed. Upper Saddle River, New Jersey, Prentice-Hall.

Zimmer-Faust, R. K.; Spanier, E. 1987: Gregariousness and sociability in spiny lobsters: implications for den habitation. *Journal of Experimental Marine Biology and Ecology 105*: 57–71.

Zimmer-Faust, R. K.; Tyre, J. E.; Case, J. F. 1985: Chemical attraction in the spiny lobster, *Panulirus interruptus* (Randall), and its probable ecological significance. *Biological Bulletin 169*: 106–118.